

AquaSense Phase 1

Source Code and Project Documentation

Automatic Touchless Water Dispenser using Arduino Uno, HC-SR04, I2C LCD, and two SG90 servo motors.

Project Files

File	Purpose
AquaSense_Phase1.ino	Main Arduino program for sensing, display control, and valve actuation.
README.md	Project setup, libraries, operating logic, upload instructions, and troubleshooting.
Pin_Configuration.txt	Hardware wiring and Arduino pin assignments.

AquaSense_Phase1.ino

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <Servo.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);
Servo servo1;
Servo servo2;

const uint8_t TRIG_PIN = 9;
const uint8_t ECHO_PIN = 10;
const uint8_t SERV01_PIN = 6;
const uint8_t SERV02_PIN = 7;

const float DETECT_DISTANCE_CM = 15.0;
const uint8_t OPEN_ANGLE = 90;
const uint8_t CLOSE_ANGLE = 0;
const unsigned long DISPENSING_TIME_MS = 3000;
const unsigned long THANK_YOU_TIME_MS = 2000;
const unsigned long LOOP_DELAY_MS = 200;
const unsigned long ECHO_TIMEOUT_US = 30000;

float getDistanceCm();
void displayStandby(float distanceCm);
void displayDispensing();
void displayThankYou();
void closeValve();
void openValve();

void setup()
{
  Serial.begin(9600);
  lcd.init();
  lcd.backlight();

  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  digitalWrite(TRIG_PIN, LOW);

  servo1.attach(SERV01_PIN);
  servo2.attach(SERV02_PIN);
  closeValve();

  lcd.setCursor(0, 0);
  lcd.print("Automatic Water");
  lcd.setCursor(0, 1);
  lcd.print("Dispenser");
  delay(2000);
  lcd.clear();
}

void loop()
{
  const float distanceCm = getDistanceCm();

  Serial.print("Distance: ");
  if (distanceCm <= 0.0) {
    Serial.println("No echo");
  } else {
```

```

Serial.print(distanceCm, 1);
Serial.println(" cm");
}

displayStandby(distanceCm);

if (distanceCm > 0.0 && distanceCm <= DETECT_DISTANCE_CM) {
displayDispensing();
openValve();
delay(DISPENSING_TIME_MS);

closeValve();
displayThankYou();
delay(THANK_YOU_TIME_MS);
lcd.clear();
}

delay(LOOP_DELAY_MS);
}

float getDistanceCm()
{
digitalWrite(TRIG_PIN, LOW);
delayMicroseconds(2);
digitalWrite(TRIG_PIN, HIGH);
delayMicroseconds(10);
digitalWrite(TRIG_PIN, LOW);

const unsigned long durationUs = pulseIn(ECHO_PIN, HIGH, ECHO_TIMEOUT_US);
if (durationUs == 0) return -1.0;
return (durationUs * 0.0343) / 2.0;
}

void displayStandby(float distanceCm)
{
lcd.setCursor(0, 0);
lcd.print("Place Hand ");
lcd.setCursor(0, 1);
lcd.print("Dist:");

if (distanceCm <= 0.0) {
lcd.print("--- cm ");
} else {
lcd.print(distanceCm, 1);
lcd.print(" cm ");
}
}

void displayDispensing()
{
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Dispensing");
lcd.setCursor(0, 1);
lcd.print("Water...");
}

void displayThankYou()
{
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Thank You");
lcd.setCursor(0, 1);
lcd.print("Visit Again");
}

void openValve()
{
servo1.write(OPEN_ANGLE);
servo2.write(OPEN_ANGLE);
}

void closeValve()
{
servo1.write(CLOSE_ANGLE);
servo2.write(CLOSE_ANGLE);
}

```

README.md

AquaSense Phase 1

Automatic touchless water dispenser using an Arduino Uno, HC-SR04 ultrasonic sensor, I2C 16x2 LCD, and two SG90 servo motors.

Libraries

- Wire.h – I2C communication support.
- LiquidCrystal_I2C.h – I2C LCD control.
- Servo.h – SG90 servo control.

Arduino Board

- Board: Arduino Uno
- Serial baud rate: 9600
- Logic level: 5 V

Pin Configuration

- HC-SR04 TRIG -> D9
- HC-SR04 ECHO -> D10
- Servo 1 signal -> D6
- Servo 2 signal -> D7
- I2C LCD SDA -> A4
- I2C LCD SCL -> A5

System Settings

- LCD address: 0x27
- LCD size: 16x2
- Detection distance: 15 cm
- Servo open angle: 90 degrees
- Servo closed angle: 0 degrees
- Dispensing time: 3000 ms (3 seconds)

Operation

1. The Arduino initializes the LCD, sensor, and servos.
2. Both servos move to the closed position.
3. The HC-SR04 measures the hand distance.
4. If the distance is between 0 and 15 cm, dispensing begins.
5. Both servos move to 90 degrees and open the valve.
6. Water flows for 3 seconds.
7. Both servos return to 0 degrees and close the valve.
8. The LCD displays Thank You / Visit Again.
9. The system returns to standby.

Upload

Select Arduino Uno, choose the correct COM port, install LiquidCrystal_I2C if required, verify the code, and upload it using the Arduino IDE.

Troubleshooting

If the LCD is blank, check wiring and try address 0x3F. If the servos jitter or reset the Arduino, use a separate regulated servo supply with a common ground. Adjust the servo angles if the mechanical valve does not fully open or close.

Safety

Use a protected 18650 battery pack and suitable charger. Keep electronics and battery connections away from water.

Pin_Configuration.txt

AquaSense Phase 1 - Pin Configuration

Controller: Arduino Uno

HC-SR04:

- VCC -> 5V
- GND -> GND
- TRIG -> D9
- ECHO -> D10

I2C LCD 16x2:

- VCC -> 5V or regulated supply
- GND -> GND
- SDA -> A4
- SCL -> A5
- I2C address -> 0x27

SG90 Servo 1:

- Signal -> D6
- VCC -> regulated servo supply
- GND -> common GND

SG90 Servo 2:

- Signal -> D7

VCC -> regulated servo supply
GND -> common GND

Settings:
Detection distance -> 15 cm
Open angle -> 90 degrees
Closed angle -> 0 degrees
Dispensing time -> 3 seconds

Important: Use a common ground and protect all electronics from water.